Aerospace Manufacturing Core
Study Guide

"AMCSG-206"

Tools & Processes

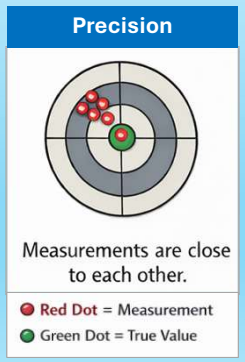
Precision Instruments

Intro to Precision Instruments

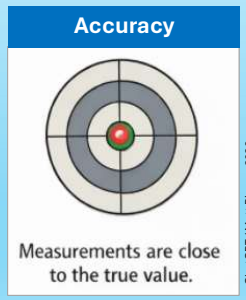
Text from Google AI

Fundamental Concepts

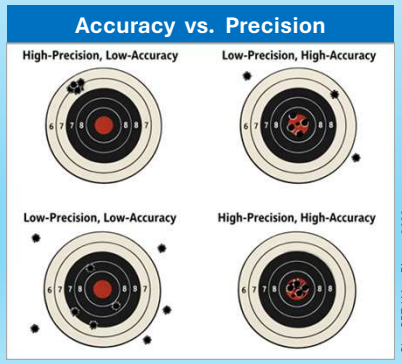
- **Precision** = Repeatability - How consistent is tool.
- **Accuracy** = Trueness - How close Measured value is to Real value.
- “Accuracy vs. Precision” - Must be both in aircraft manufacturing



ChatGPT / Matt Pierce, 2026



ChatGPT / Matt Pierce, 2026



ChatGPT / Matt Pierce, 2026

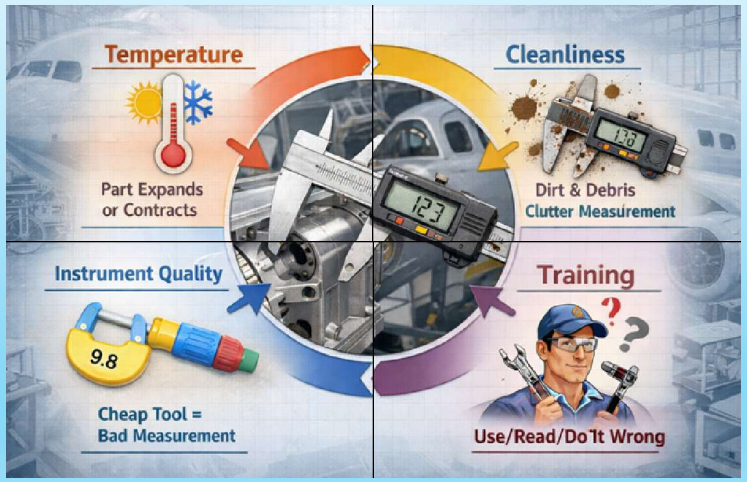
- Google AI: “write a pictorial guide to the general use of precision instruments in aerospace manufacturing. Define "Precision and Accuracy". Define "Environment" (especially temperature expanding metal). Define "cleanliness" (dirty parts hard to measure). Define "accuracy" and "calibration" and use of "masters". Define "measurement tool discrimination". Define "resolution". Define how to take a measurement from the "reference point" to the "measurement point" to get the "measurement distance“”

- ChatGPT: “make a diagram showing the difference between (1) Precision (repeatability), (2) Accuracy (Trueness), and (3) Accuracy vs. Precision. Context is aircraft manufacturing precision tools like a caliper or a micrometer.“
  + Remove the picture of a caliper. Remove the picture of a micrometer. Make the bullseye a green dot everywhere
  + In picture #2, make the red dot appear dead center in the green dot. Make the title for picture #3 the same red color as pictures #1 an #2.
  + Remove the red titles. Make the red dot in picture #2 same size as red dots in picture #1.
-

Text from ChatGPT

Factors Influencing Precision & Accuracy

- **Temperature**:
  - Metal parts expand or contract
- **Cleanliness**:
  - Dirty part or gauge = bad measure
  - Use solvents, wipe clean
- **Instrument Quality**:
  - Cheap tool = bad measure
- **Training & Usage**:
  - Use it wrong, Read it wrong, Do it wrong = bad measure



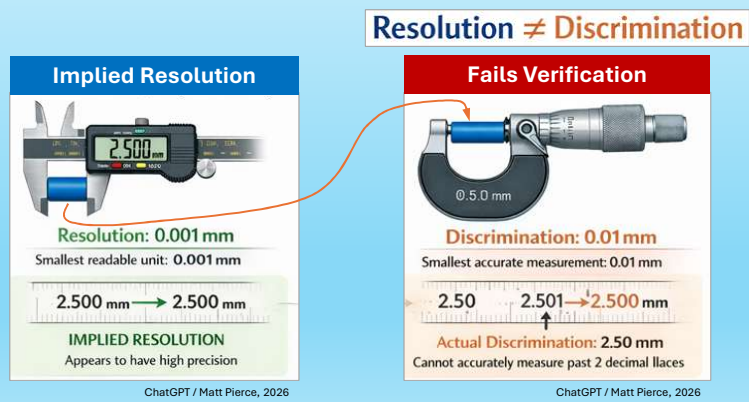
ChatGPT / Matt Pierce, 2026

ChatGPT: Make a diagram showing 4 factors that influence precision and accuracy of aircraft manufacturing precision tools: 1. Temperature (part expands or contracts) 2. Cleanliness (dirt and debris clutter measurement) 3. Instrument Quality (cheap tool = bad measurement) 4. Training (use it wrong or read it wrong = bad measurement)
+ Remove the blue title.
+ Change title from "Use It Wrong or Read it Wrong" to "Use/Read/Do It Wrong".
+ Change tool in picture "Instrument Quality" to a really cheap "Micrometer" that is the cheapest model out there with few options and poor quality.
+ Make the micrometer in the lower left picture made out of plastic and like a kids toy.

Text from ChatGPT

Resolution vs. Discrimination

- **Resolution** = Smallest unit of measure readable from instrument
- **Discrimination** = Smallest value instrument can accurately measure



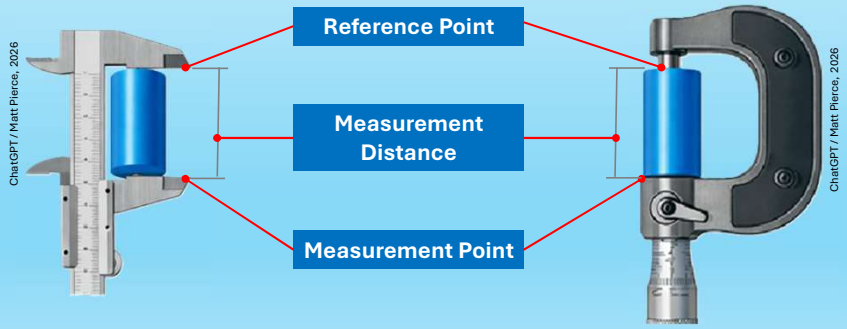
- ChatGPT: make a diagram using a precision instrument like a caliper or micrometer that shows the difference between "resolution" (smallest readable unit of measure) and "discrimination" (smallest value instrument can measure)
  + redo the diagram using same instruments, but instead emphasize that an instrument may imply it has great resolution, but its actual discrimination (ability to accurately measure) is less than the resolution.
  + show the micrometer instrument on the right side measuring the same part as the caliper on the left side. color the part being measured by each tool in bright blue.

Text from ChatGPT

Measurement Definitions

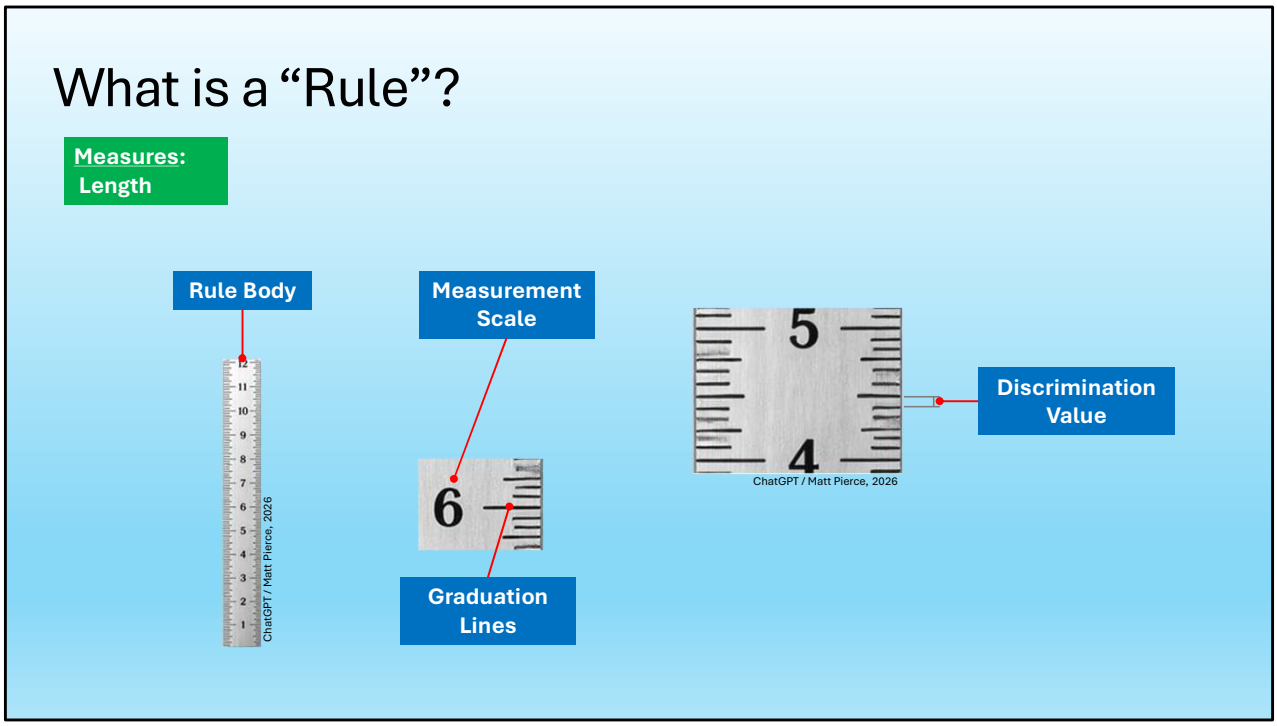
Measurements generally have three parts:

- **Reference Point** (from static / non-moving part of instrument)
- **Measurement Point** (from moving part of instrument)
- **Measurement Distance** (the difference)

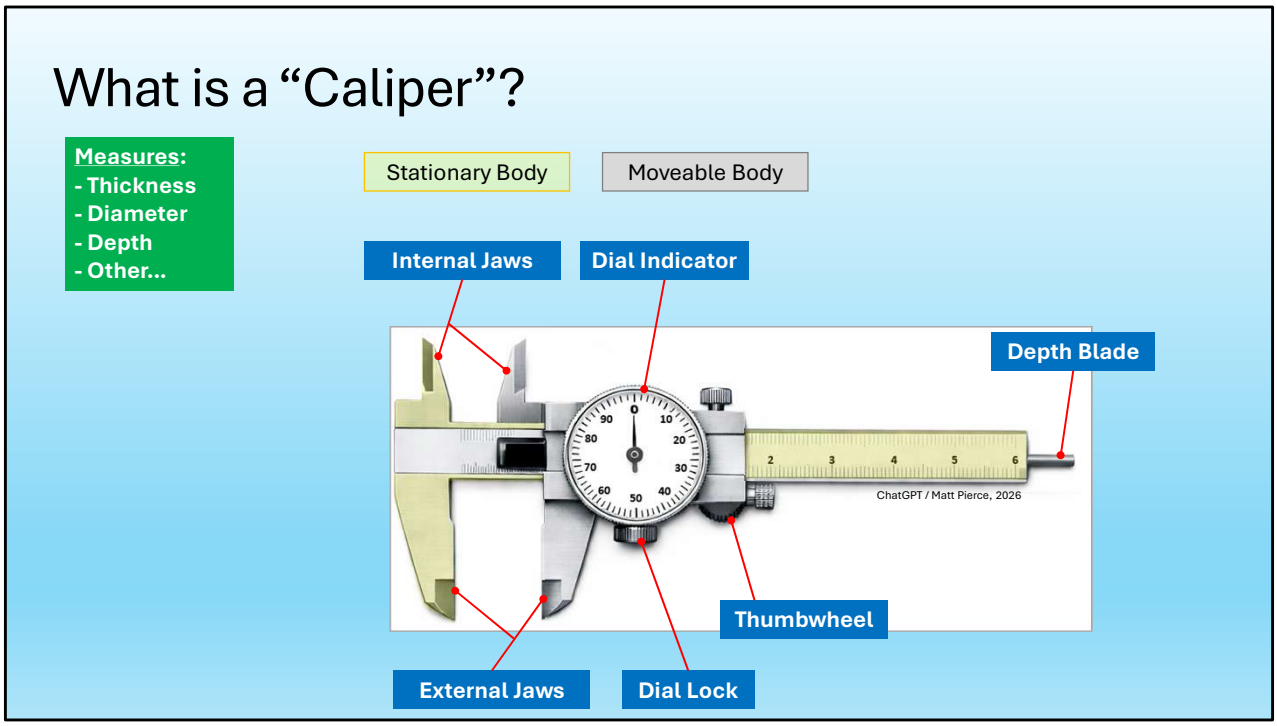


- ChatGPT: make diagram comparing a micrometer measuring a bright blue cylinder on the right, and a caliper measuring the same blue cylinder on the left.

Types of Precision Instruments



- ChatGPT: make a diagram of a metal rule (not a ruler). make it standing vertical. make the digits increment from 1" to 12" in one inch increments.



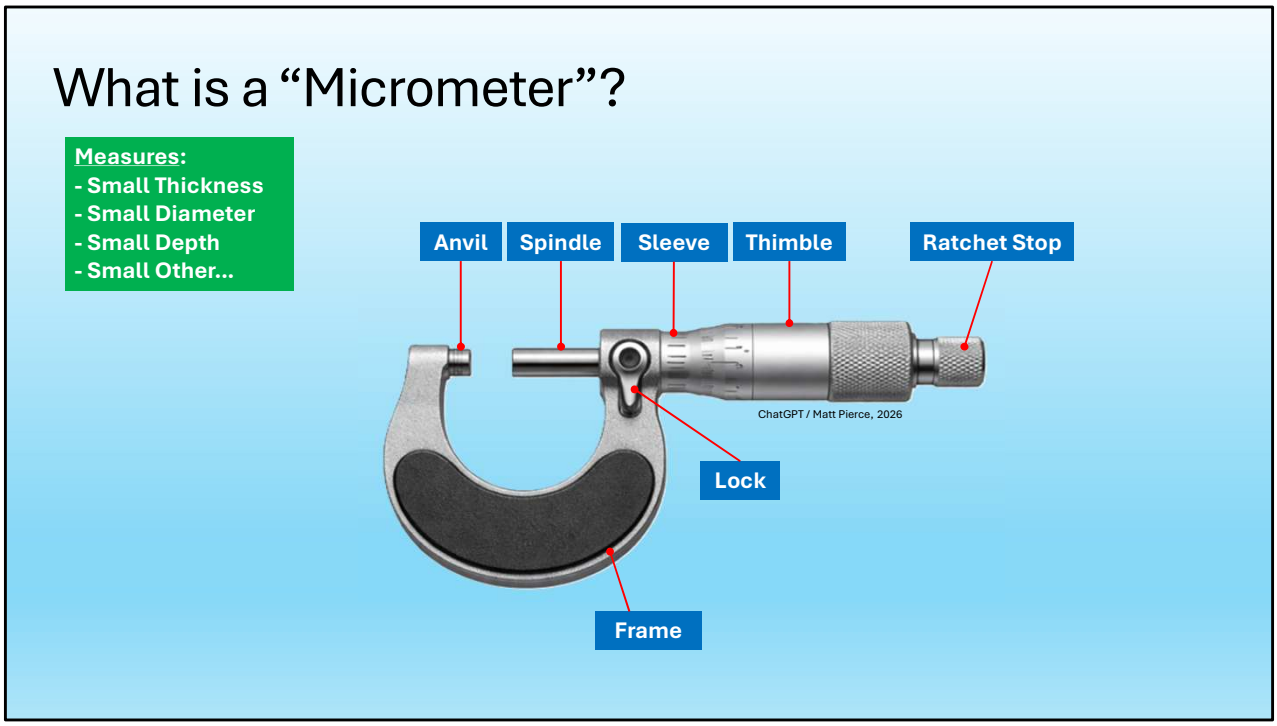
- ChatGPT: make a single diagram showing the parts of a dial caliper
  + Use picture #3. Remove metrics measurements everywhere.
  + Remove all labels and arrows.
  + Add a blowup image showing the value measured, including the vernier scale, the main scale and how the reading is taken
  + make the yellow dial background all white
  + make the dial indicator go from 0 to 100
  + change the dial indicator numeric values to the following in clockwise order starting from 0 at the top: 10, 20, 30, 40, 50, 60, 70, 80, 90. Use only 10 numbers (not 11 as is currently shown)

Text from Google AI

How to Use a Caliper

1. **Zero Out Instrument** - Close jaws completely; needle points to "0". If not: loosen dial lock, rotate dial face to zero, re-tighten dial lock.
2. **Read Main Beam (Inches)** - Look at last numbered digit sliding jaw passed. This is whole inch measurement.
3. **Read Main Beam (Tenths)** - Count smaller hash marks passed after whole inch. Each hash = 0.100” (1/10th”)
4. **Read Dial (Thousandths)** - Read number on dial face. Each mark is 0.001” (1/1000th”)
5. **Calculate Total** - Add the values together (Inches + Tenths + Dial). For example, if the beams shows 1.3” and the dial reads 56, the measurement is 1.3 + 0.056 = 1.356”

- Google AI: “how to read a dial caliper”



- ChatGPT: make a single diagram showing the parts of a micrometer
  + Remove all labels, show only the micrometer for picture #2

List from Google AI

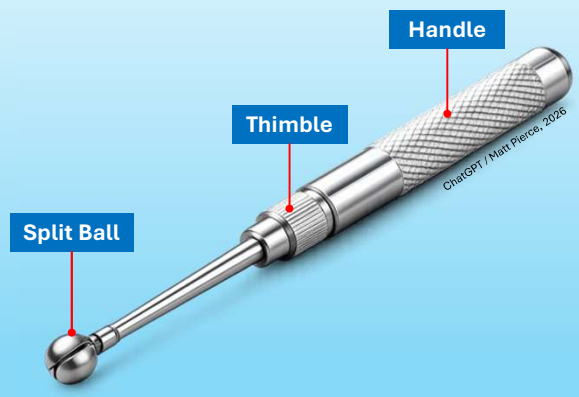
How to Use a Micrometer

1. **Zero Out Instrument** - Close spindle completely using ratchet stop to avoid over-tightening; if sleeve not align to 0 use supplied spanner wrench to reset to 0
2. **Read Main Sleeve (Tenths)** - Identify last visible numbered digit. This is 0.100" (1/10" ... so a "3" would be 3/10" or 0.300").
3. **Read Main Sleeve (25-Thousandths)** - Count smaller vertical marks past the number. Each hash = 0.025" (25/1000th")
4. **Read Thimble (Thousandths)** - Read number on thimble aligning with horizontal index line of sleeve. Each mark is 0.001" (1/1000th")
5. **Calculate Total** - Add the values together. For example, Sleeve shows "2" (0.200") + 3 small lines (0.075") + Thimble shows "12" (0.012")
   Thus Total = 0.200 + 0.075 + 0.012 = 0.287"

- Google AI: "how to read a micrometer"
- Google AI: "how to zero out a micrometer"

What is a “Split Ball Gauge”?

Measures:
- Small Hole Diameter
- Slot Width
- Recess Depth



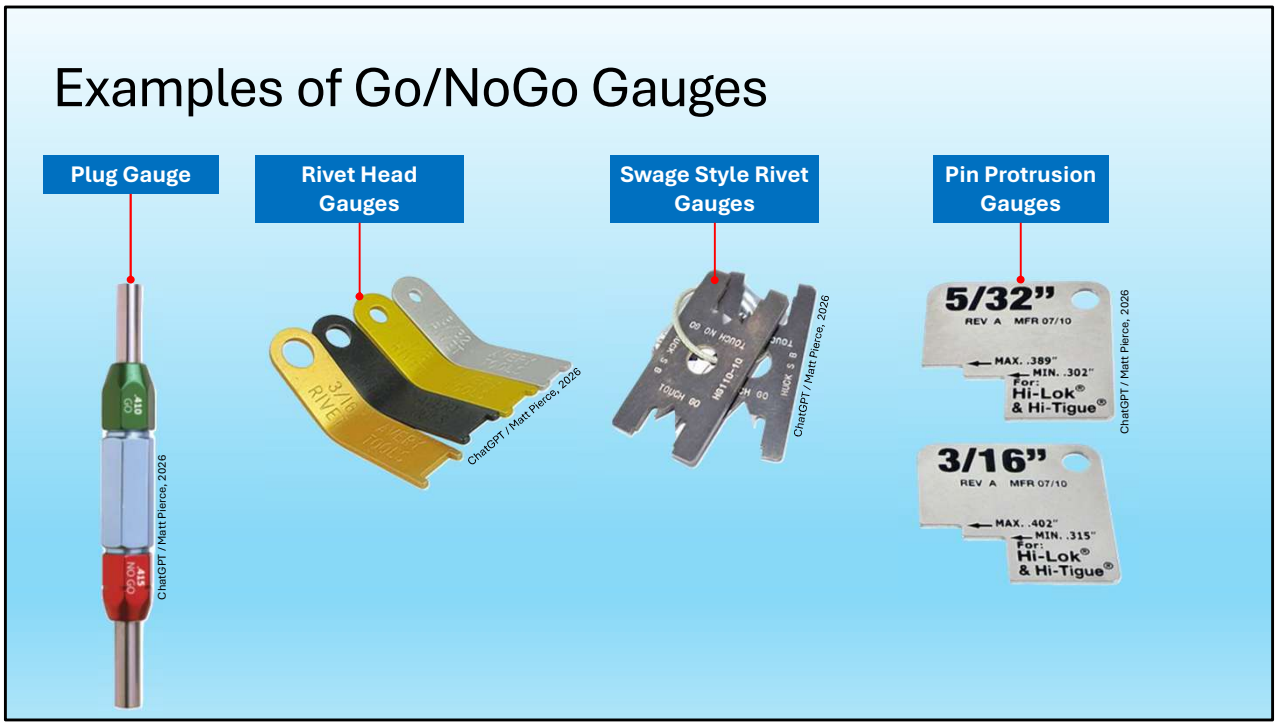
- ChatGPT: Make a photo of a split ball gauge used to measure small hole diameter
  + remove all the backdrop

Text from Google AI

How to Use a Split Ball Gauge

1. **Select & Prepare** - Choose correct size gauge for hole. Clean it and hole.
2. **Insert & Adjust** - Insert contracted gauge into hole. Rotate Thimble to expand ball until it makes light contact with hole walls.
3. **Locate Max Dia (Rocking)** - Gently rock gauge back-and-forth, side-by-side, up-and-down the hole to locate the maximum diameter
4. **Lock and Remove** - If available, lock the gauge. Remove gauge.
5. **Measure** - Use a micrometer to measure the split ball head

- Google AI: “how to read a micrometer”
- Google AI: “how to zero out a micrometer”



- ChatGPT: show me a picture of Go/NoGo Plug Gauge. Heavily manually edited.
- ChatGPT: show me a picture of Rivet Inspection Guage. Took two pictures, one for rivet head gauge set, other for swage style rivet gauges
- ChatGPT: show me a picture of a pin protrusion gauge

Q&A

Practice Q&A's (Q1-Q8)

Q1: Name for repeatability; how consistently a tool measures?

A: Precision

Q2: Name for how close the measured value is to the real value?

A: Accuracy

Q3: Y/N: Can a precision measurement tool be accurate, but not precise?

A: Yes. Even a broken analog watch is accurate twice a day (but not at all precise)

Q4: What factor makes a metal part expand or contract?

A: Temperature

Q5: How do you fix a dirty part or gauge so that precision and accuracy are optimal?

A: Wipe clean with a solvent

Q6: How can you avoid using, reading, or doing precision measurements wrong?

A: Proper training and practice

Q7: Name of the smallest visible unit of measure readable on the precision instrument?

A: Resolution

Q8: Name of the smallest unit of measure the precision instrument can accurately measure, regardless of what is readable?

A: Discrimination

Practice Q&A's (Q9-Q16)

Q9: Name of tip of non-moving part of precision instrument from which measurement starts?

A: Reference Point

Q10: Name of tip of moving point on a precision instrument where measurement ends?

A: Measurement Point

Q11: Name of the distance between the reference and measurement points?

A: Measurement Distance

Q12: Name of a precision instrument used to draw straight lines and measure distance typically up to 12"?

A: Rule

Q13: What is the difference between a Rule and a Ruler?

A: Rule edge starts at 0". Ruler has about 1/16" gap before the 0" mark starts (for kids to wear away)

Q14: Name of precision instrument that uses jaws to measure up to 6" in distance, with a blade to measure depth too?

A: Caliper

Q15: What is the first step you should do every time you take a caliper measurement to ensure accuracy?

A: Zero-Out Caliper

Q16: Name of precision instrument that uses an anvil and spindle to measure distance?

A: Micrometer

Practice Q&A's (Q17-Q)

Q17: Name of precision instrument that goes inside a hole and captures the diameter to be measured by a micrometer?

A: Split Ball Gauge

Q18: Another name for fixed (non-variable) precision instruments?

A: Go/NoGo Gauges

Thank You

Matt Pierce - mpierce710006@gmail.com - linkedin.com/in/mpiercead576/

https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg